

PADHAISHURU

GATE Preparation Platform

GATE EC

Electronics & Communication Engineering

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 22 questions (34%)

Moderate: 37 questions (57%)

Difficult: 6 questions (9%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE EC — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

2M [-0.66]

[General Aptitude]

For a real number $x > 1$, $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \log_2 x + \log_3 x + \log_4 x = 1$. The value of x is

- (A) 4
- (B) 12
- (C) 24
- (D) 36

Q2.

2M [-0.66]

[General Aptitude]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Q3.

2M [no -ve]

[General Aptitude]

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room. If S and Q are single children, and the room allocations shown below are acceptable to all, PR TS Q PQ RT S then, which one of the given options is the siblings?

Enter your answer here

Q4.

2M [-0.66]

[General Aptitude]

If ` denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored _____ weary]. Which one of the given options is appropriate to fill the blank?

- (A) jaded
- (B) baffled
- (C) dead
- (D) worsted

Q5.

2M [no -ve]

[General Aptitude]

If train 100m crosses pole in 10s, speed:

Enter your answer here

Q6.

2M [-0.66]

[General Aptitude]

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II. Group-I: Abuse Insult Ridicule Group-II: _____ Praise Appreciate

- (A) Extol
- (B) Prize
- (C) Appropriate
- (D) Espouse

Q7.

2M [-0.66]

[General Aptitude]

The 12 musical notes are given as $\sharp$, $\flat$, $\sharp$, $\flat$, $\sharp$, $\flat$, $\sharp$, $\flat$, $\sharp$, $\flat$, $\sharp$, and $\flat$. Frequency of each note is 122 times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $\sharp$ and $\flat$ is:

- (A) 62
- (B) 2
- (C) 42
- (D) 2

Q8.

2M [no -ve]

[General Aptitude]

Sum of first 100 natural numbers:

Enter your answer here

Q9.

2M [no -ve]

[General Aptitude]

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration' is: Note: The figures shown are representative. 1 Iteration 0: Iteration 1: 1
1 Length of each segment: 3 3 1 1 Iteration 2: 9 Length of each segment: 9

Enter your answer here

Q10.

2M [-0.66]

[General Aptitude]

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

- (A) will be
- (B) can be
- (C) am going to be
- (D) could have been

Q11.

2M [no -ve]

[General Aptitude]

Choose synonym of 'trivial':

Enter your answer here

Q12.

2M [-0.66]

[Engineering Mathematics]

$i^4 =$:

- (A) 0
- (B) 1
- (C) -1
- (D) i

Q13.

2M [-0.66]

*[Engineering Mathematics]*Integral of $1/x$ dx:

- (A) x
- (B) $\log(x)+C$
- (C) $1/x^2$
- (D) $x^2/2$

Q14.

1M [-0.33]

[Engineering Mathematics] $P(A \cap B) = ?$

- (A) $P(A)+P(B)$
- (B) $P(A)+P(B)-P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q15.

2M [-0.66]

[Network, Signals & Systems]

For a causal discrete-time LTI system with transfer function $H(z) = (2z^2 + 3z + 13) / (z^2 - 13z + 13)$ - which of the following statements is/are true?

- (A) The system is stable.
- (B) The system is a minimum phase system.
- (C) The initial value of the impulse response is 2.
- (D) The final value of the impulse response is 0.

Q16.

2M [no -ve]

[Network, Signals & Systems]

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____. $t=0$ 3H S 5A 2 4 4

- (A) A
- (B) B
- (C) C
- (D) D

Q17.

2M [-0.66]

[Network, Signals & Systems]

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} /m^3$ at 300 K. If the electron and hole mobilities are $0.15 m^2/Vs$ and $0.05 m^2/Vs$, respectively, then the intrinsic resistivity of the semiconductor (in k.m) at 300 K is _____. (Charge of an electron = $1.6 \times 10^{-19} C$.)

- (A) 1.65
- (B) 1.25
- (C) 0.85
- (D) 1.95

Q18.

2M [-0.66]

[Network, Signals & Systems]

Consider the following series: (i) $\sum_{n=1}^{\infty} 1$ (ii) $\sum_{n=1}^{\infty} 1$ (+1) (iii) $\sum_{n=1}^{\infty} 1$! Choose the correct option.

- (A) Only (ii) converges
- (B) Only (ii) and (iii) converge
- (C) Only (iii) converges
- (D) All three converge

Q19.

2M [no -ve]

[Network, Signals & Systems]

The propagation delay of the 2-to-1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as 0 ns. If S is set to 1 then the output Y is _____.

Enter your answer here

Q20.

2M [-0.66]

[Network, Signals & Systems]

An ideal p-n junction germanium diode has a reverse saturation current of 10^{-10} A at 300 K. The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____. (Consider the Boltzmann constant $B = 1.38 \times 10^{-23}$ J/K and the charge of an electron $= 1.6 \times 10^{-19}$ C.)

- (A) A
- (B) B
- (C) C
- (D) D

Q21.

2M [-0.66]

[Network, Signals & Systems]

An NMOS transistor operating in the linear region has $I_D = 5$ A at $V_{GS} = 0.1$ V. Keeping constant, V_{GS} is increased to 1.5 V. Given that $\mu_n C_{ox} = 50$ A/V², the transconductance at the new operating point (in A/V, rounded off to two decimal places) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q22.

2M [-0.66]

[Network, Signals & Systems]

Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $\alpha > 0$. Assuming $h_2[n] = \alpha \delta[n]$, the overall system (denoted by the dashed box) is _____.

- (A) linear and time invariant
- (B) linear and time variant
- (C) nonlinear and time invariant
- (D) nonlinear and time variant

Q23.

1M [no -ve]

[Electronic Devices]

Zener diode used for:

Enter your answer here

Q24.

1M [no -ve]

[Electronic Devices]

SCR has:

*Enter your answer here***Q25.**

1M [-0.33]

[Electronic Devices]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q26.

1M [-0.33]

[Electronic Devices]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q27.

1M [-0.33]

[Analog Circuits]

RC oscillator frequency:

- (A) $f = 1/(2\pi RC)$
- (B) $f = 1/RC$
- (C) $f = 1/(4\pi RC)$
- (D) $f = 2\pi RC$

Q28.

1M [no -ve]

[Analog Circuits]

Ideal op-amp open-loop gain:

*Enter your answer here***Q29.**

1M [-0.33]

[Analog Circuits]

The input resistance of a common emitter amplifier is

- (A) $R_B \parallel R_{in}$
- (B) $R_B \parallel R_{in} \parallel R_{in}$
- (C) $R_B \parallel R_{in}$
- (D) $R_B \parallel R_{in}$

Q30.

1M [-0.33]

[Analog Circuits]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q31.

1M [-0.33]

[Analog Circuits]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q32.

1M [no -ve]

[Digital Circuits]

ROM is:

*Enter your answer here***Q33.**

1M [no -ve]

[Digital Circuits]

4-bit binary counter counts:

*Enter your answer here***Q34.**

1M [no -ve]

[Digital Circuits]

D flip-flop follows input after:

*Enter your answer here***Q35.**

1M [-0.33]

[Digital Circuits]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q36.

1M [no -ve]

[Control Systems]

Routh-Hurwitz determines:

*Enter your answer here***Q37.**

1M [-0.33]

[Control Systems]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q38.

1M [-0.33]

[Control Systems]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q39.

1M [-0.33]

[Control Systems]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q40.

1M [-0.33]

[Communication Systems]

PCM stands for:

- (A) Pulse Code Modulation
- (B) Phase Code Modulation
- (C) Pulse Carrier Modulation
- (D) None

Q41.

1M [-0.33]

[Communication Systems]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q42.

1M [-0.33]

[Communication Systems]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q43.

1M [-0.33]

[Communication Systems]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q44.

1M [-0.33]

*[Electromagnetics]*Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q45.

1M [no -ve]

[Electromagnetics]

Waveguide supports:

Enter your answer here

Q46.

1M [-0.33]

[Electromagnetics]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q47.

1M [-0.33]

[Electromagnetics]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q48.

1M [-0.33]

[Control Systems]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q49.

2M [no -ve]

[Network, Signals & Systems]

In the circuit below, the opamp is ideal. $R_1 = C R R + R C$ If the circuit is to show sustained oscillations, the respective values of 1 and the corresponding frequency of oscillation are _____.

- (A) 29 and $1/(26)$
- (B) 2 and $1/(2)$
- (C) 29 and $1/(2)$
- (D) 2 and $1/(26)$

Q50.

2M [-0.66]

[Network, Signals & Systems]

Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3 resistor are as indicated. The voltage (in Volts) at node is _____.

- (A) $20/3$
- (B) $32/3$
- (C) $22/3$
- (D) $2/3$

Q51.

2M [-0.66]

[Network, Signals & Systems]

For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____. $I_1 I_2 1 + 2$
 $\frac{1}{2} - V_2 V_1 + 8V_1 1' - - 2'$

- (A) A
- (B) B
- (C) C
- (D) D

Q52.

2M [-0.66]

[Network, Signals & Systems]

A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V. For an input sinusoidal signal with frequency 500 kHz, the maximum SNR (in dB, rounded off to two decimal places) and the data rate (in Mbps) at the output of the ADC are _____, respectively.

- (A) 61.96 and 10
- (B) 61.96 and 5
- (C) 33.36 and 10
- (D) 33.36 and 5

Q53.

2M [-0.66]

[Network, Signals & Systems]

In the circuit below, the opamp is ideal. $R_1 _ C R R + R C$ If the circuit is to show sustained oscillations, the respective values of 1 and the corresponding frequency of oscillation are _____.

- (A) 29 and $1/(26)$
- (B) 2 and $1/(2)$
- (C) 29 and $1/(2)$
- (D) 2 and $1/(26)$

Q54.

2M [no -ve]

[Network, Signals & Systems]

Let F_1 , F_2 , and F_3 be functions of (x, y, z) . Suppose that for every given pair of points and in space, the line integral $C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at and ends at B . Then which of the following is/are true?

Enter your answer here

Q55.

2M [-0.66]

[Network, Signals & Systems]

A source transmits a symbol s , taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s . Using $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$, the optimum symbol error probability is _____.

- (A) $2Q(2)$
- (B) $4Q(1)$
- (C) $2Q(1)$
- (D) $4Q(2)$

Q56.

2M [-0.66]

[Network, Signals & Systems]

In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$, $C_1 = C$. If the circuit is to show sustained oscillations, the respective values of R and the corresponding frequency of oscillation are _____.

- (A) 29 and $1/(26)$
- (B) 2 and $1/(2)$
- (C) 29 and $1/(2)$
- (D) 2 and $1/(26)$

Q57.

2M [no -ve]

[Network, Signals & Systems]

A 4-bit priority encoder has inputs 3, 2, 1, and 0 in descending order of priority. The two-bit output is generated as 00, 01, 10, and 11 corresponding to inputs 3, 2, 1, and 0, respectively. The Boolean expression of the output bit is _____.

Enter your answer here

Q58.

1M [-0.33]

[Electromagnetics]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q59.

2M [no -ve]

[Network, Signals & Systems]

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle in the complex plane is/are TRUE?

Enter your answer here

Q60.

2M [-0.66]

[Network, Signals & Systems]

Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-B, B]$, where B is in rad/sec. The signal $x(t)$ is uniformly sampled to obtain $x[n] = x(nT)$. Here, $T = (2\pi/B)$, $\delta(t)$ is the Dirac impulse, $\delta(t) > 0$, and $\delta(t) > 0$. The sampled signal $x[n] \sin(n\pi/2)$ is passed through an ideal lowpass filter $H(f)$ with cutoff frequency and passband gain 1. The output of the filter is given by _____.

- (A) $x(t)$ if $t < T$
- (B) $x(t)$ if $t < T/2$
- (C) $x(t)$ if $t < 2T$
- (D) $x(t)$ if $t < 2T/2$

Q61.

2M [-0.66]

[Network, Signals & Systems]

A uniform plane wave with electric field $E(x) = A e^{j 2 \pi x} e^{j 3 \pi y}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the + direction (A is a positive constant, a^y is the unit vector along the axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $y = 0$. The position of the first null of the total magnetic field in the air (measured from $y = 0$, in metres) is _____.

- (A) -3
- (B) -3
- (C) -6
- (D) -3

Q62.

2M [no -ve]

[Network, Signals & Systems]

A digital communication system transmits through a noiseless bandlimited channel $[-B, B]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum b[n]x(t - nT)$ where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = nT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

Enter your answer here

Q63.

2M [-0.66]

[Network, Signals & Systems]

Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $x(t)$ is the input signal. The differential equations governing the system are given by $\dot{x}_1(t) = -x_1(t) + x_2(t)$, $\dot{x}_2(t) = x_1(t) - 2x_2(t) + x_3(t)$, $\dot{x}_3(t) = x_2(t) - x_3(t) + x(t)$. Which of the following statements is/are TRUE?

- (A) The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs
- (B) There exists a bounded input such that at least one of the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ is unbounded
- (C) There exists a bounded input such that the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded
- (D) The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded for all bounded inputs

Q64.

1M [-0.33]

[Network, Signals & Systems]

Two identical sheets A and B, of dimensions 24 cm $\times$ 16 cm, can be folded into half using two distinct operations, FO1 or FO2. In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge. If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

- (A) 14:11
- (B) 11:14
- (C) 18:11
- (D) 11:18

Q65.

1M [-0.33]

[Network, Signals & Systems]

Four identical cylindrical chalk-sticks, each of radius = 0.5 cm and length = 10 cm, are bound tightly together using a duct tape as shown in the following figure. The width of the duct tape is equal to the length of the chalk-stick. The area (in cm²) of the duct tape required to wrap the bundle of chalk-sticks once, is

- (A) $20(4 + \pi)$
- (B) $20(8 + \pi)$
- (C) $10(8 + \pi)$
- (D) $10(4 + \pi)$